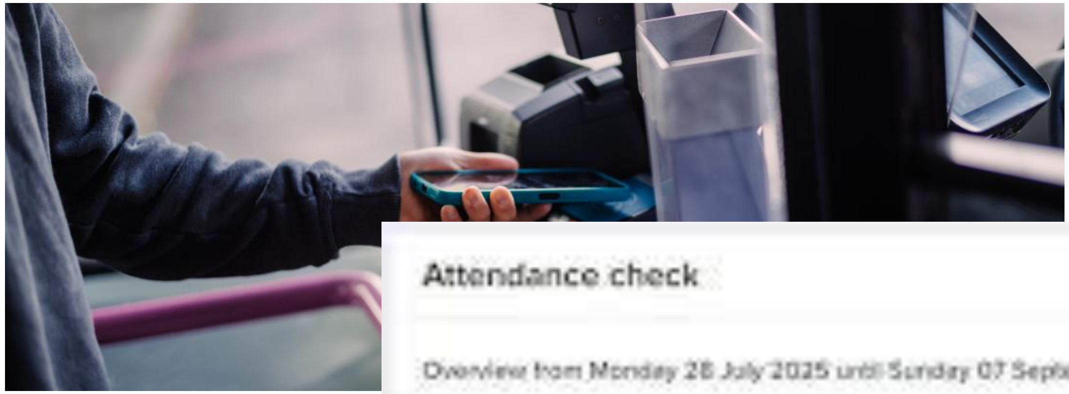
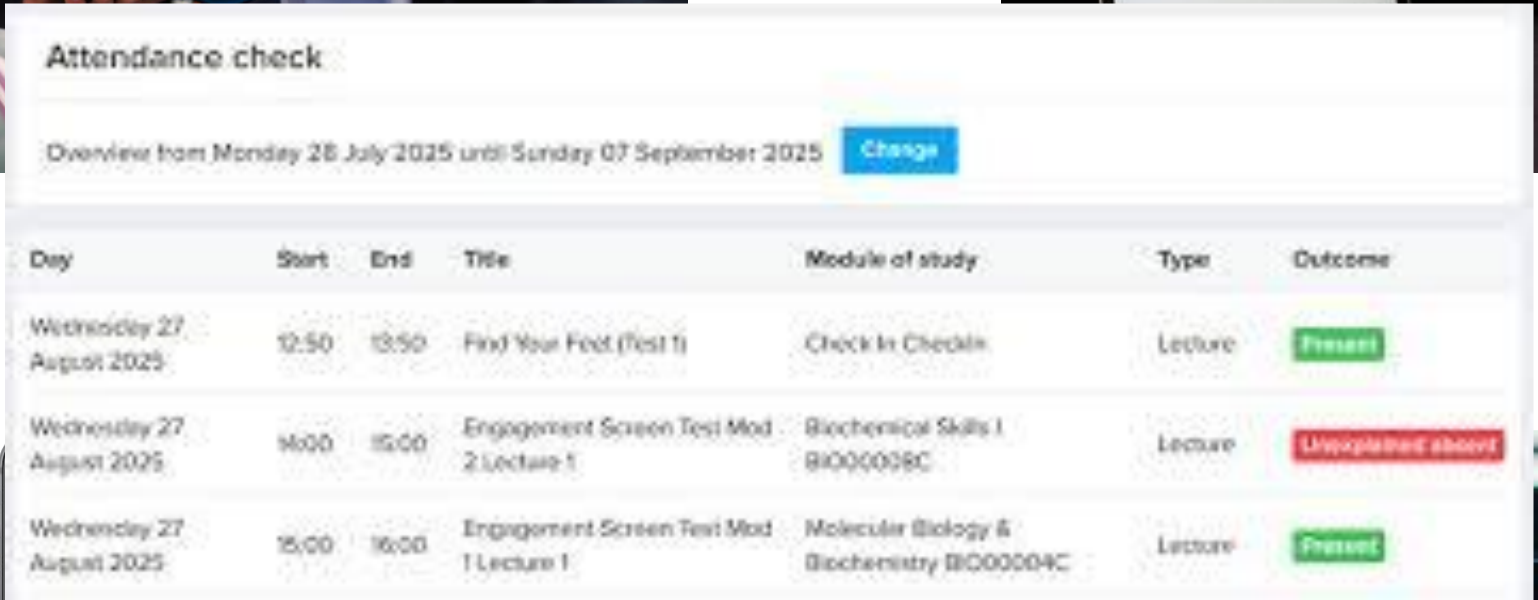
Introduction to Big Data
Week 10





Attendance check

Overview from Monday 28 July 2025 until Sunday 07 September 2025 Change

Day	Start	End	Title	Module of study	Type	Outcome
Wednesday 27 August 2025	12:50	13:50	Find Your Feet (Test 1)	Check In CheckIn	Lecture	Present
Wednesday 27 August 2025	14:00	15:00	Engagement Screen Test Mod 2 Lecture 1	Biochemical Skills I BIO00008C	Lecture	Unexplained absence
Wednesday 27 August 2025	15:00	16:00	Engagement Screen Test Mod 1 Lecture 1	Molecular Biology & Biochemistry BIO00004C	Lecture	Present

Introduction to Big Data

- **Big Data** refers to extremely large, complex datasets that cannot be easily managed, processed, or analysed using traditional methods.

- Big Data has become **essential for competitive advantage**, enabling businesses to predict market trends and patterns, and personalise customer experiences, and make informed decisions.

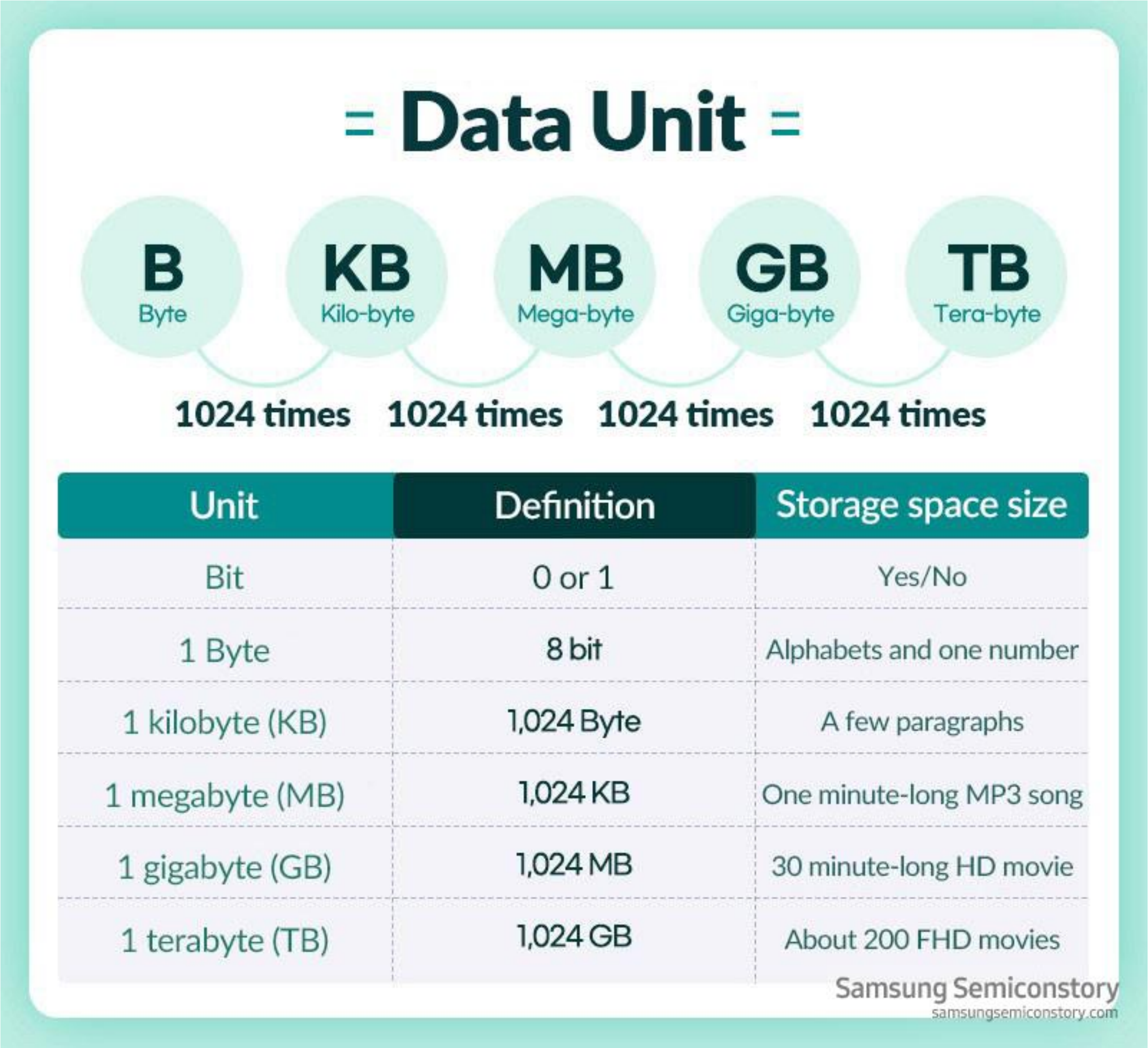
- The quantity of data generated is growing exponentially, including data generated by:

  - Retail e-commerce databases
  - User-interactions with websites and mobile apps
  - Usage of logistics, transportation systems, financial and health care
  - Social media data
  - Location data (e.g. GPS-generated)
  - Internet of Things (IoT) data generated
  - New forms of scientific data (e.g. human genome analysis)
  - …

How Big is Big?

Big data isn't defined by a specific byte size but by its scale, exceeding the capacity of **traditional software** to capture, curate, manage, and process within a reasonable time.

While data was once measured in gigabytes, it now typically involves terabytes, petabytes, or even exabytes, a volume so large and complex that it requires **distributed computing** and specialized tools for effective use.

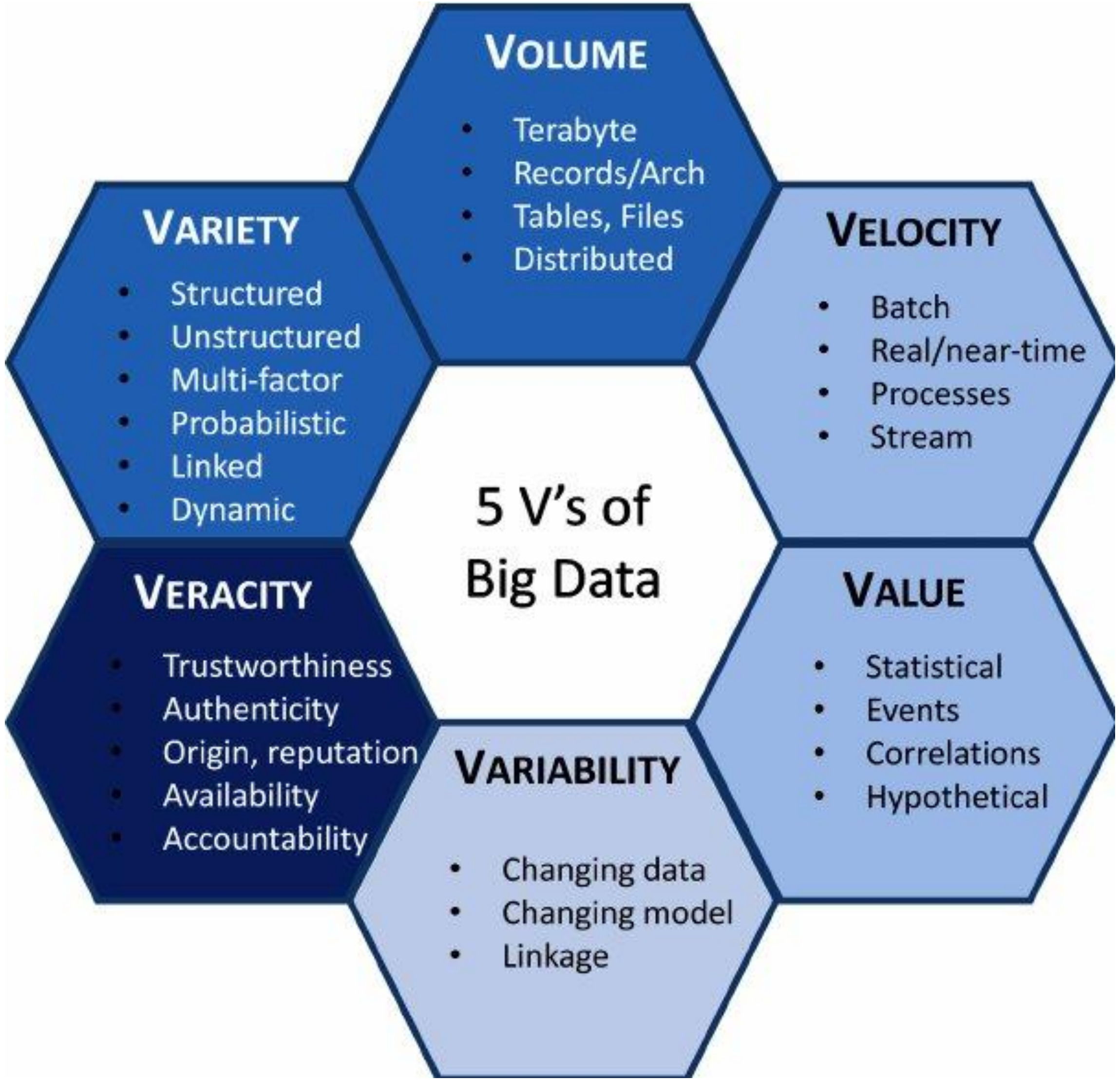


Unit	Definition	Storage space size
Bit	0 or 1	Yes/No
1 Byte	8 bit	Alphabets and one number
1 kilobyte (KB)	1,024 Byte	A few paragraphs
1 megabyte (MB)	1,024 KB	One minute-long MP3 song
1 gigabyte (GB)	1,024 MB	30 minute-long HD movie
1 terabyte (TB)	1,024 GB	About 200 FHD movies

5 V’s of Big Data

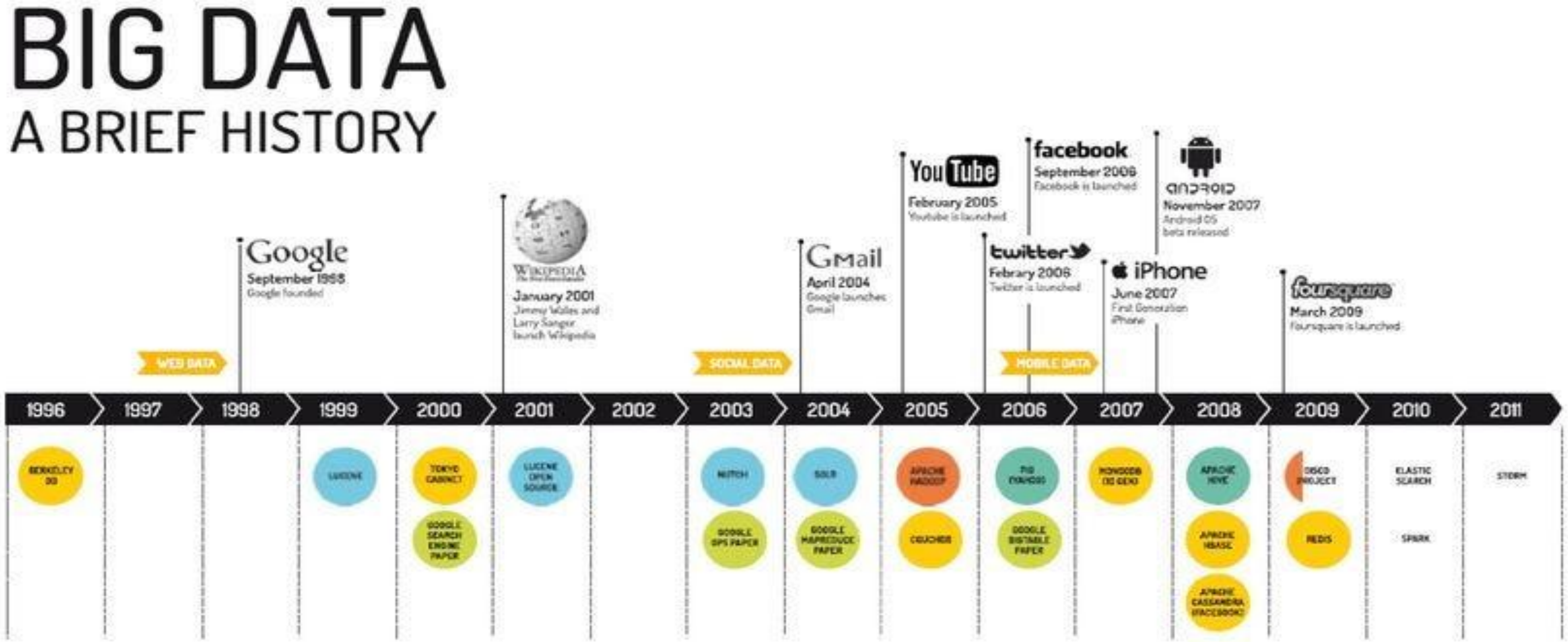
- It is often described by five (or six) key characteristics (the **5 V**s):

  - **Volume** – massive amounts of data
  - **Velocity** – data generated very quickly
  - **Variety** – different types and formats (text, images, video, etc.)
  - **Veracity** – uncertainty and quality issues in data
  - **Value** – potential to generate insights and economic value
    - **Variability** – Changing data, model.

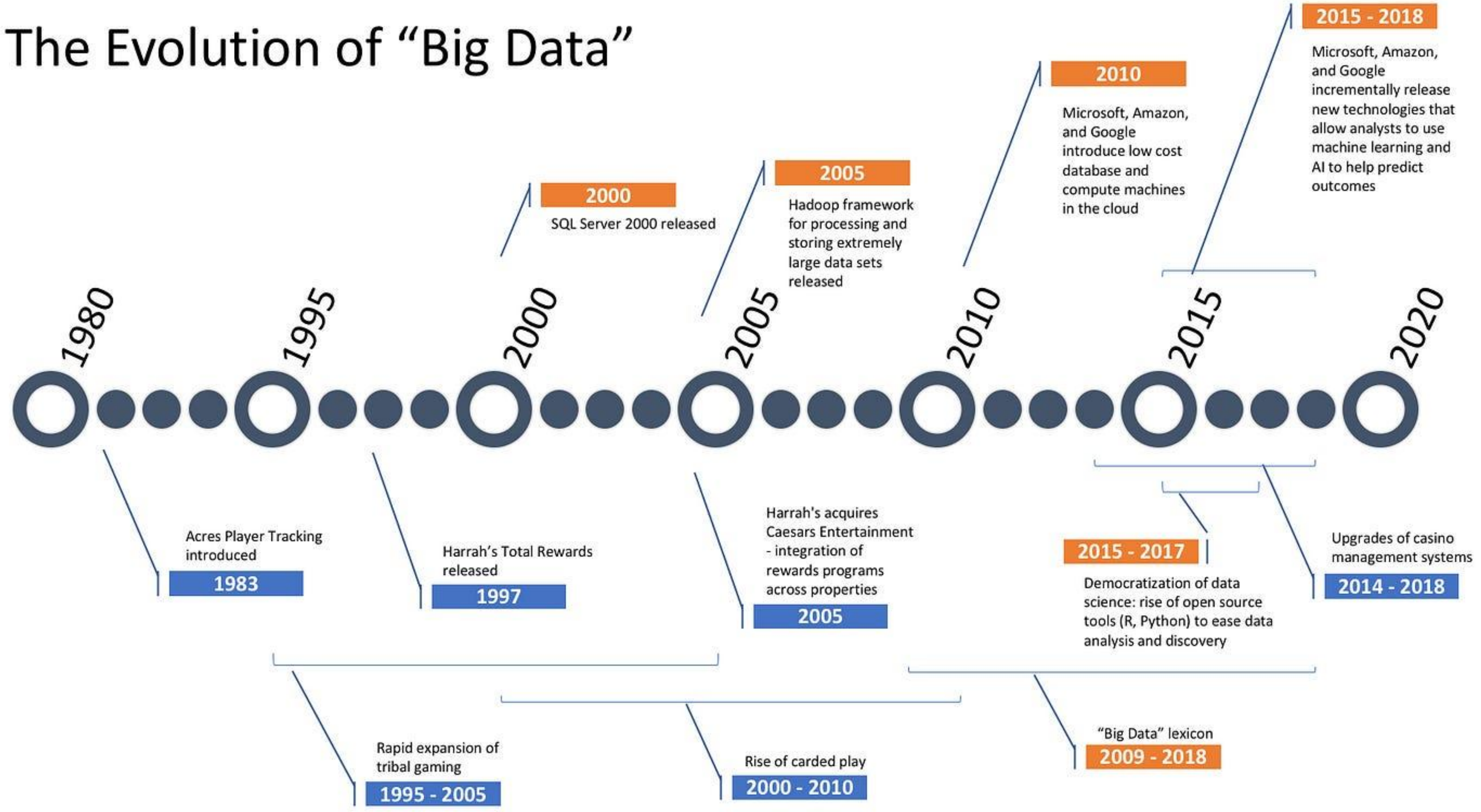


Big Data

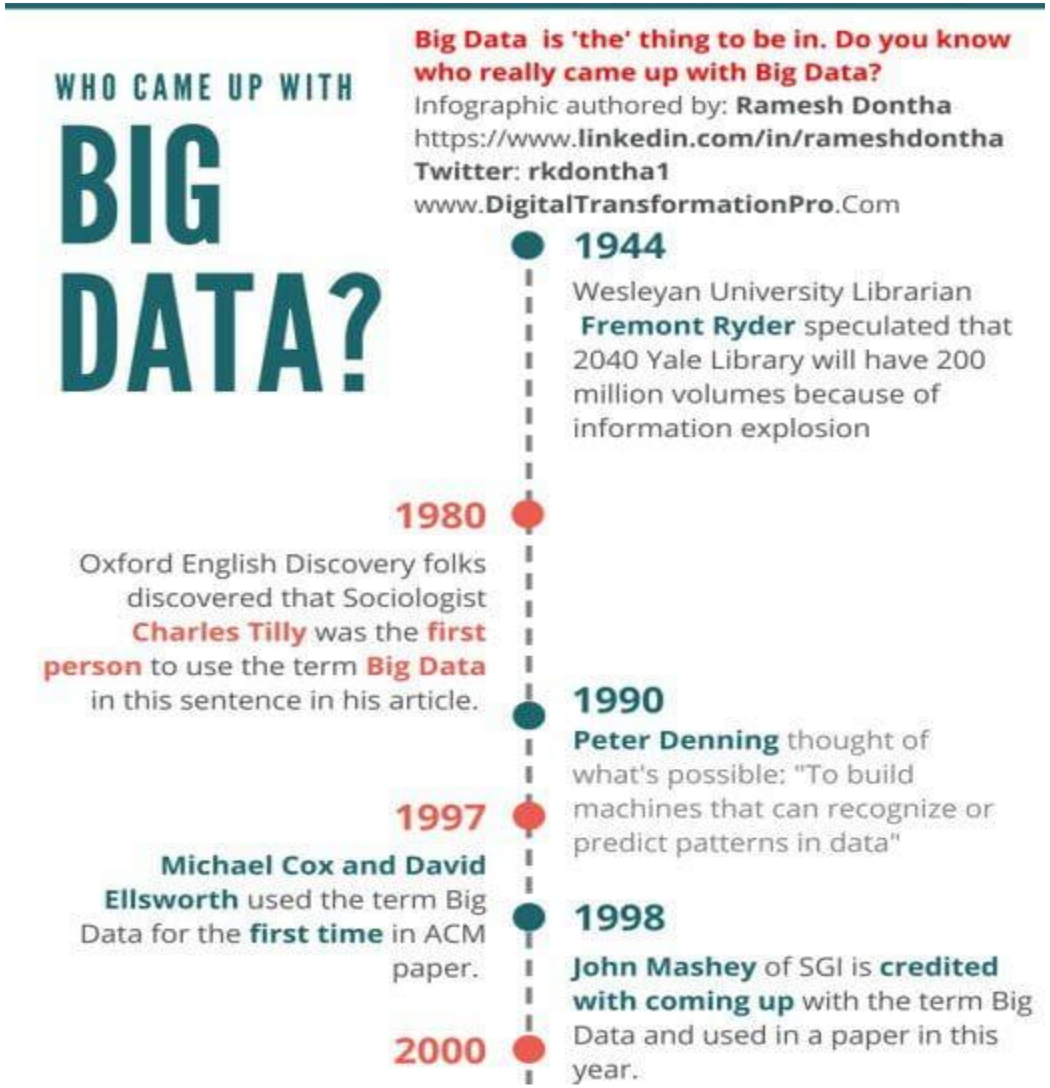
Big Data – a Brief History



Big Data Technology



A brief history



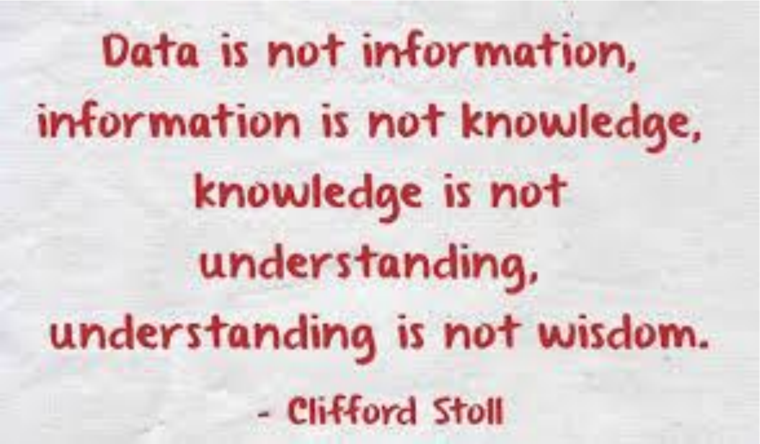
What is Data Analytics?

- **Data Analytics** means using techniques, algorithms, and technologies to discover patterns, trends, and insights from data — especially from large and complex datasets.

- It includes **descriptive** (*what happened*?), **diagnostic** (*why*?), **predictive** (*what will happen*?), and **prescriptive** (*what should we do*?) analysis.

- In Big Data Analytics, we need **special tools** because:

  - The data is too big,
  - Arrives too fast,
  - Comes in too many different forms (text, video, sensor signals, etc.).



Big Data Analytics

Category	Traditional Analysis	Big Data Analytics
Data Characteristics	• Small, clean datasets	• Massive, messy, fast (high-velocity) data
Typical Techniques	• ANOVA, chi-square tests, t-tests • Classical regression • Time-series decomposition	• Machine learning and AI • Streaming analytics, real-time dashboards
Models / Algorithms	• Linear / logistic regression • ARIMA / ETS • Clustering	• decision trees, random forests • Neural networks (Deep learning) • Transformer
Workflow & Speed	• Results after careful modeling • Batch, scheduled analysis	• Instant or dynamic analysis • Continuous / real-time pipelines
Outputs	• Static reports, tables, charts	• Interactive dashboards, live KPIs
Common Tools	• Excel • SPSS, Stata, Eviews	• Hadoop, Spark, cloud platforms • R, MATLAB, Python (rich libraries)

Need for distributed computing

Modern business data is too large and too fast for one computer.

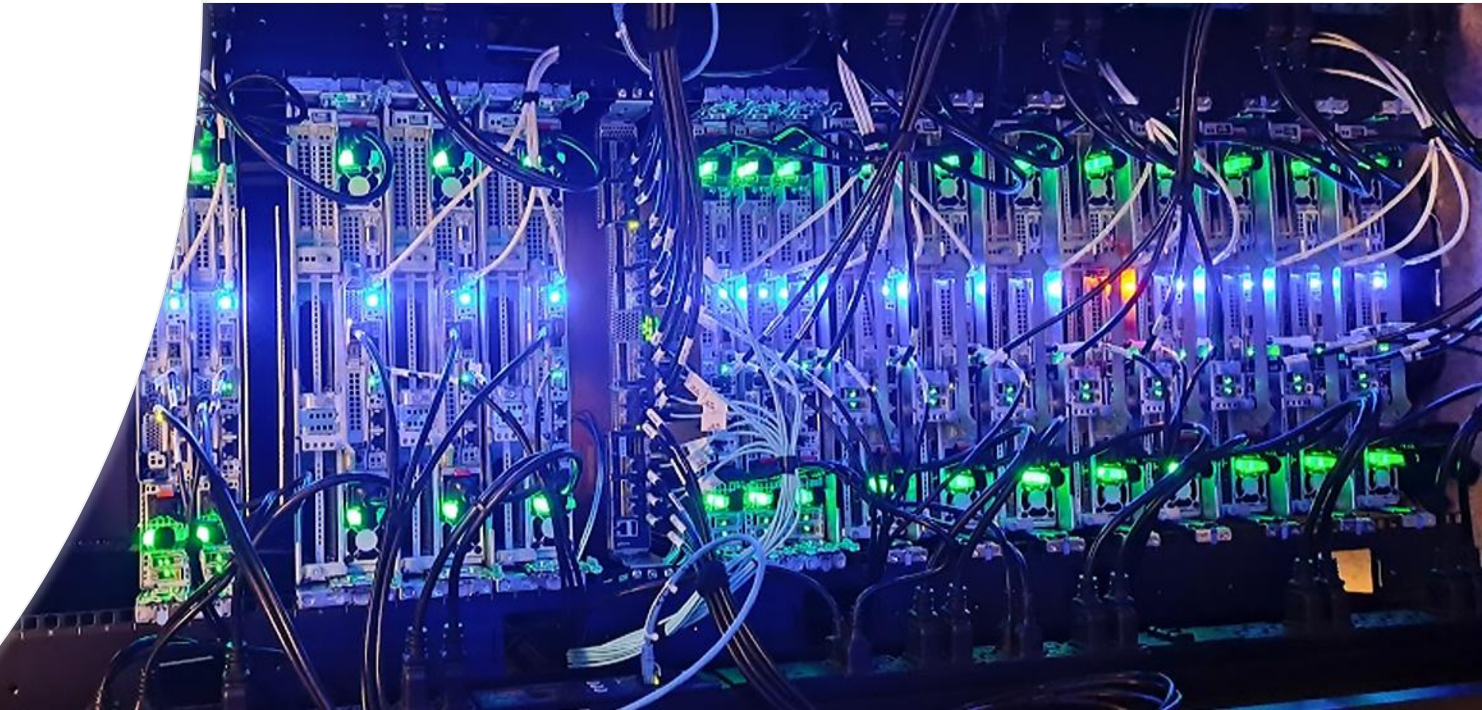
- Companies collect millions or billions of data points
  - Transactions, clicks, text, images, messages
- One computer has limited memory and processing speed
- Processing everything on one machine would be:
  - Too slow
  - Too expensive
  - Sometimes impossible

Solution: Distributed computing

- Split the data across many computers
- Each computer processes a small part
- All computers work at the same time
- Results are combined into one answer

Computer Cluster at UoY

- The University has invested £2.5m to build a new high-performance computing facility (Viking2).
- 134 compute nodes: each with 96 CPU cores per node
- It is the world’s first data centre to be made from wood and runs on 100% renewable energy; 75% from hydropower and 25% from wind power.



How Big Data calculations actually work

When companies work with Big Data, the problem is the **size of the data**.

- Big Data is too large to fit on one computer
- One machine cannot process it fast enough
- Using a bigger computer is expensive and inefficient

The solution:
Split the data into smaller parts and process them separately. (**divide and conquer**)

In Big Data systems:

- Data is stored across many computers, not one
- Each computer works on its own part of the data
- All computers work at the same time (this is called **parallel computation**)
- Instead of moving all data to one place:
- Each computer sends back small results
- These results are then combined

Example of computing the mean using divide & conquer

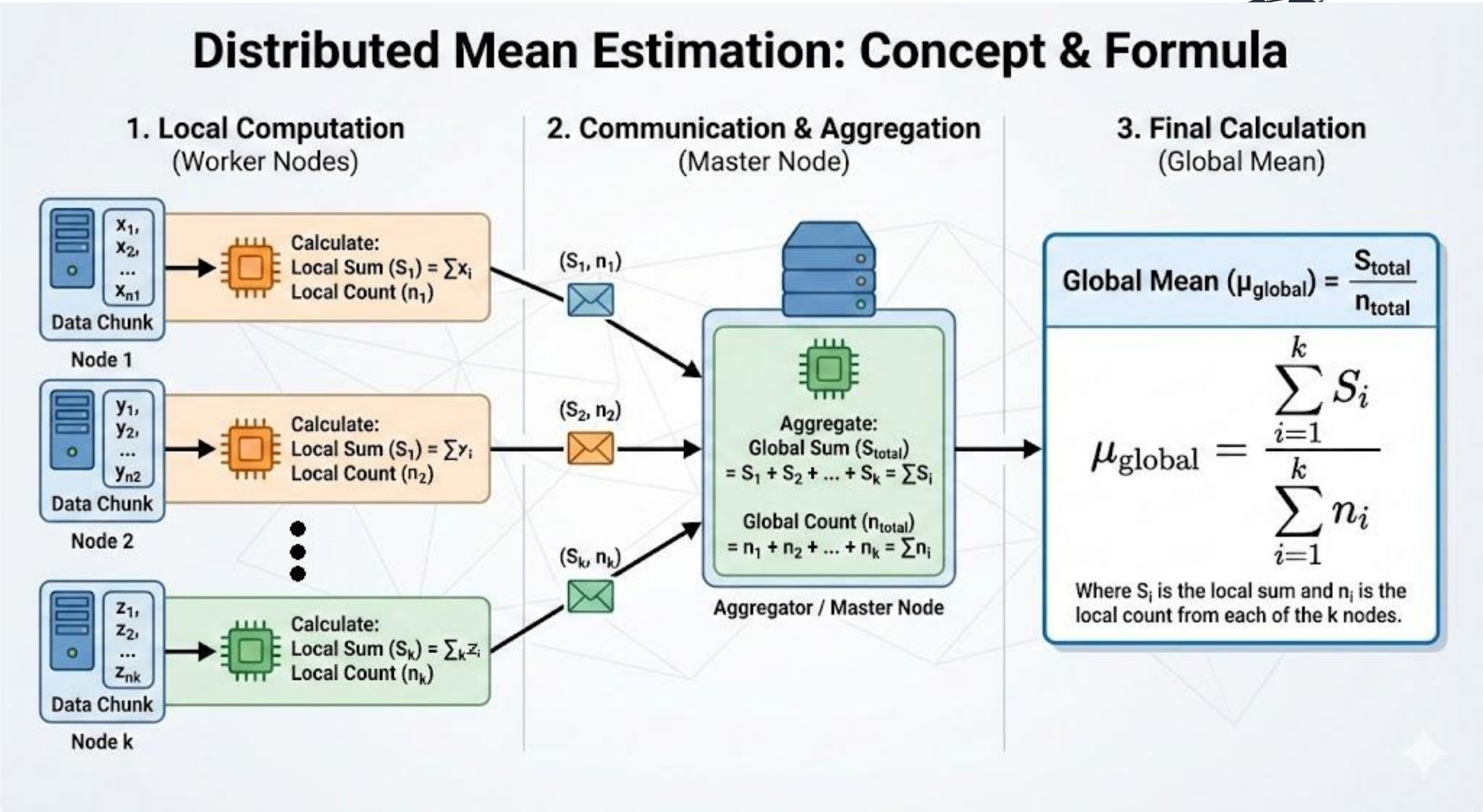
Suppose sales data is split across several computers.

Each computer calculates:

- The **sum** of its sales values
- The **number** of sales it holds
- These results are sent to a central computer.

The central computer:

- Adds all the sums together
- Adds all the counts together
- Divides total sum by total count



Distributed Mean Estimation: Concept & Formula
1. Local Computation
(Worker Nodes)
2. Communication & Aggregation
(Master Node)
3. Final Calculation
(Global Mean)
x_1, x_2, ..., x_{n1}
Data Chunk
Node 1
Calculate:
Local Sum (S_1) = $\sum x_i$
Local Count (n_1)
y_1, y_2, ..., y_{n2}
Data Chunk
Node 2
Calculate:
Local Sum (S_1) = $\sum y_i$
Local Count (n_2)
z_1, z_2, ..., z_{nk}
Data Chunk
Node k
Calculate:
Local Sum (S_k) = $\sum_k z_i$
Local Count (n_k)
(S_1, n_1)
(S_2, n_2)
(S_k, n_k)
Aggregate:
Global Sum (S_{total})
= $S_1 + S_2 + ... + S_k = \sum S_i$
Global Count (n_{total})
= $n_1 + n_2 + ... + n_k = \sum n_i$
Aggregator / Master Node
Global Mean (μ_{global}) = $\frac{S_{total}}{n_{total}}$
$$\mu_{\text{global}} = \frac{\sum_{i=1}^{k} S_i}{\sum_{i=1}^{k} n_i}$$
Where S_i is the local sum and n_i is the local count from each of the k nodes.

Example of computing the variance using divide & conquer

To calculate variance in Big Data:

Each computer calculates:

- Local counts
- Local sum
- Local sum of squared values

These numbers are then combined to get the overall variance.

Why Some Statistics Are Easy and Others Are Hard

Statistics that are easy in Big Data:

- Total sales
- Number of customers
- Mean (average)
- Variance

These work well because:

- We can **add results together**
- Order of the data does not matter

Statistics that are harder:

- Median
- Percentiles (e.g. top 10%)

These are harder because:

- We need to **sort the data**
- Sorting large datasets across many computers is slow

Example of computing the mean using online updating

Sometimes data arrives **continuously**, for example:

- Live website traffic
- Financial transactions
- Sensor data
- Instead of storing all data, we update the mean as data arrives.

We keep:

- The current mean
- The number of observations
- Each new value slightly adjusts the mean.

$$\bar{X}_{n+1} = \frac{1}{n+1}\left(\sum_{i=1}^{n} x_i + x_{n+1}\right) = \frac{n}{n+1}\bar{X}_n + \frac{1}{n+1}x_{n+1}$$

Why this is useful:

- No need to store all data
- Works in real time
- Ideal for fast-moving business data

Big Data in Business and Management

- Demand & Supply Optimization
  - **Walmart** uses Big Data to **optimise supply chain** management and store inventory levels, analysing real-time data from point-of-sale systems, social media, and weather forecasts to predict product demand.
  - **Uber**'s entire service model is built on Big Data. From **pricing** to **optimising pick-up and drop-off points** and **predicting peak demand times**, Big Data analytics drives operational efficiency and customer satisfaction.
  - **Airbnb** uses Big Data to **adjust pricing in real-time based on changes in demand and supply conditions**. This dynamic pricing model helps hosts maximise their earnings while providing competitive pricing for guests.

- Store/Network & Product Strategy
  - **Starbucks** uses Microsoft Power BI to analyse and visualise data collected from various sources, including POS systems and customer feedback, to **optimise store locations and product offerings**.

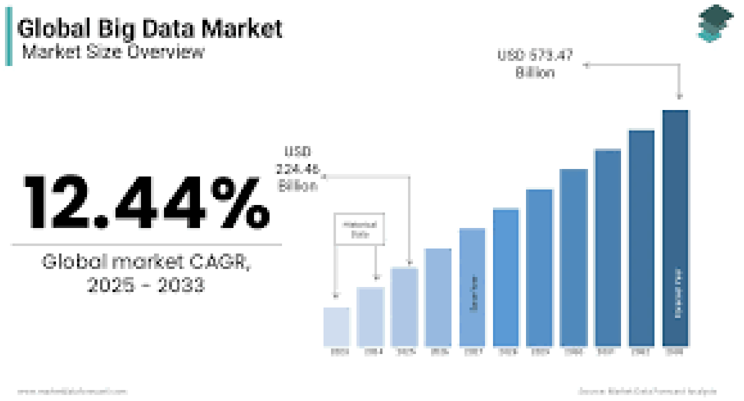
Big Data in Business and Management

- Personalization & Recommendation
  - **Amazon** is known for its sophisticated use of Big Data, Amazon analyses customer behaviour data to **personalise shopping experiences**, resulting in significantly higher sales through targeted product recommendations.
  - **American Express** uses Big Data analytics to **analyse transaction data and predict customer loyalty**, enabling targeted marketing strategies that improve customer retention rates.
- Marketing
  - **Nike** uses Big Data to **monitor real-time customer activity** and feedback on its various platforms to quickly adapt its products and marketing strategies, enhancing customer engagement and product innovation.
  - **Netflix** uses Big Data to drive its **content recommendation** engine, helping to maintain high user engagement by personalising viewing suggestions based on individual watching habits and preferences. This data-driven strategy has also guided Netflix in its decisions about which original content to produce.

Skills in Big Data

- **Data Literacy**
  - Understand what data means and how it’s collected.
  - Read and interpret charts, tables, and dashboards confidently.
- **Basic Statistics**
  - Know key ideas like averages, correlation, and simple regression.
  - Use data to support decisions rather than rely on guesses.
- **Programming (Python or R)**
  - Learn basic coding to organize, clean, and analyze data.
  - Use tools to create visualizations and simple models.
- **Critical Thinking**
  - Ask the right business questions and connect data to real problems.
  - Turn numbers into insights that guide smart decisions.
- **Communication Skills**
  - Explain findings clearly to people without a technical background.
  - Create simple visuals and tell a story with data.

- **Work with Large Language Model**



The global big data market was worth USD 199.63 billion in 2024. The global market size is expected to grow from USD 224.46 billion in 2025 to USD 573.47 billion by 2033.

What Does an LLM Actually Do?

When an LLM writes a sentence, it:

- Looks at the words already written
- Calculates the probability of the next word
- Chooses a word with a high probability

Example： A bigram model

The model looks at pairs of consecutive words.
-Given one word, what is the probability of the next word?

Suppose our "language model" is trained on these 5 sentences:
I like business analytics!
I like data!
I like marketing!
Students like business!
Companies like business!

Step 1: From the sentences, we extract word pairs (bi-grams):
(I, like) → appears **3 times**
(Students, like) → appears **1 time**
(Companies, like) → appears **1 time**
(like, business) → appears **3 times**
(like, data) → appears **1 time**
(like, marketing) → appears **1 time**
Output: I like

Step 2: After "like", we see:

Next word	Count
business	3
data	1
marketing	1

After "like", the most likely next word is "business"
P(business| like) = 3/5
Output: I like business

Step 3: After "business", the most likely next word/token is "!"

Output: I like business!

Careers in Big Data

Company	**Industry**	**Example Roles**
Amazon	E-commerce, Cloud Computing	Data Analyst, Business Intelligence Engineer
JP Morgan Chase	Finance and Banking	Risk Analyst, Quantitative Analyst
Deloitte	Consulting and Audit	Data Analytics Consultant, Risk Advisory Analyst
Netflix	Entertainment	Consumer Insights Analyst, Data Scientist
Tesla	Automotive and Energy	Business Intelligence Specialist, AI/ML Engineer
HSBC	Banking	Data Analyst, Financial Modelling Expert

Role	What They Do
Data Analyst	Clean, process, and find insights from datasets
Business Intelligence Analyst	Use data to support business decisions
Data Scientist	Build advanced models to predict future trends
Risk Analyst	Predict and prevent financial or operational risks
Marketing Analyst	Analyse customer behaviour to boost sales

Data protection and UK GDPR

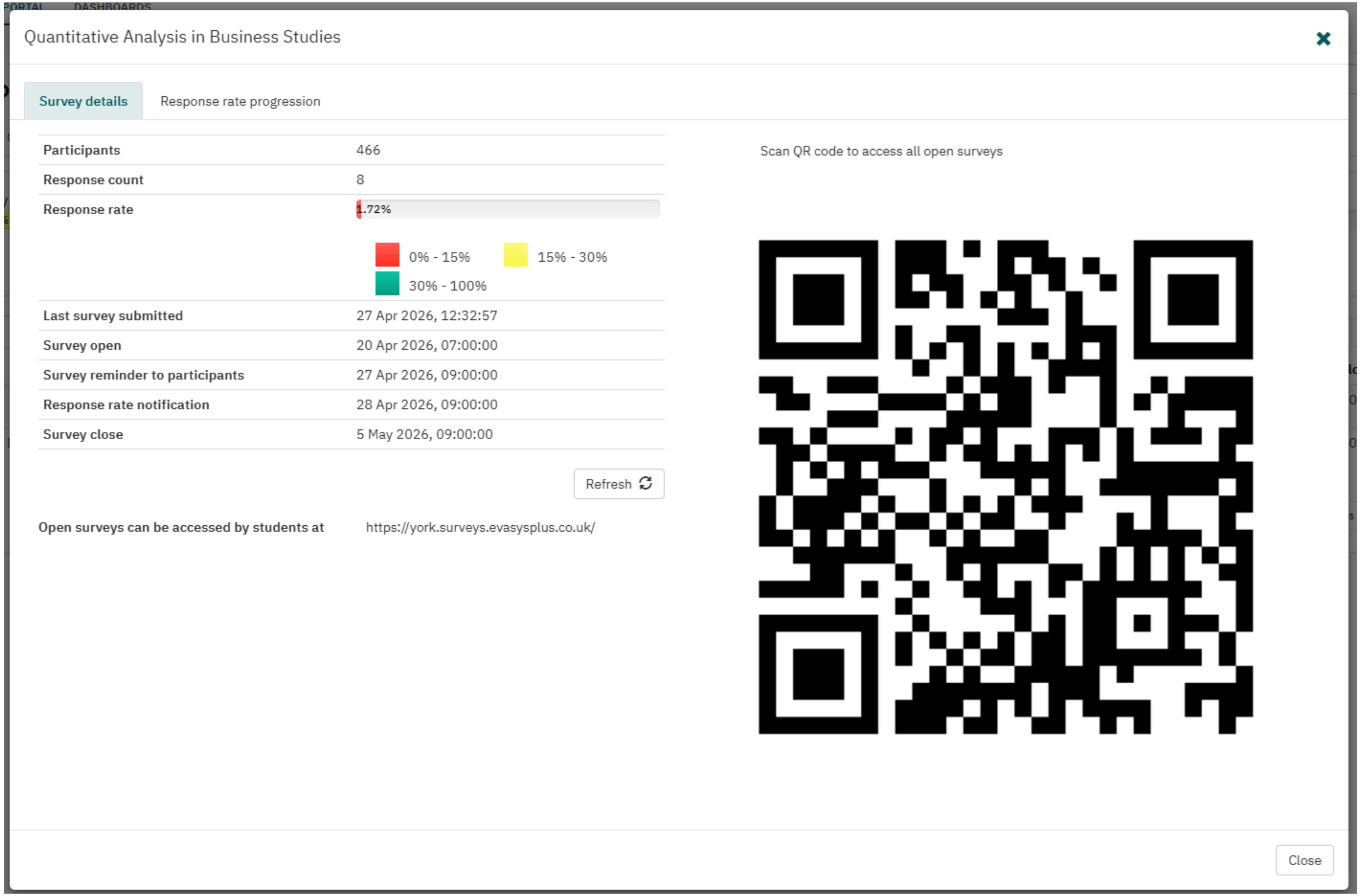
- Useful links:

[Data Protection - Records Management and Information Governance, University of York]

[Economics, Law, Management, Politics and Sociology Ethics Committee (ELMPS) - Research Centre for Social Sciences, University of York]

The UK GDPR (General Data Protection Regulation) is a law that sets rules for how personal data should be collected, stored, and used, ensuring individuals' privacy rights are protected in the UK.

(before) Next Week Plan



Next Week

- Revision for the exam
- Solving mock exam